DX699 AI for Leaders

Capstone Milestone One:

Identifying a Problem and Relevant Datasets

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1. Description or summary of the nature and content of each dataset

Dataset	Size	Source	Industry	Unit of observation
1. SPY options snapshot	9,646 x 23	Historic Options Dataset (Dubach, 2025)	Finance	One listed option contract, Dec 31, 2024
2. China EV sales	38,806 x 10	CPCA via Kaggle (felixzhao)	Environment	One vehicle model in one month, Jan 2018-Apr 2024
3. Spotify tracks	114,000 x 20	Ultimate Spotify Tracks DB	Entertainment	One track (audio features + genre)
4. TMDB movies	45,466 x 24	The Movies Dataset	Entertainment	One film (budget, revenue, ratings)
5. FINRA short interest, 30 Sep 2026	22,581 x 14	FINRA Equity Short Interest	Finance	One stock, settlement date 30 Sep 2026

Table 1. The five datasets: size, source, industry, and unit of observation.

Dataset 1: SPY Equity Options Chain Snapshot (Finance)

A single-day cross-section of the SPY options chain on December 31, 2024 (spot 586.08), hand-extracted from a public historic U.S. equity options dataset (Dubach, 2025): 9,646 contracts (4,823 calls, 4,823 puts) across 35 expirations and 362 strikes in 23 columns, covering contract

identifiers (strike, call/put, expiration), market observables (bid, ask, last, sizes, volume, open interest), and model-derived outputs (implied volatility, delta, gamma, theta, vega, rho, moneyness, in-the-money flag). Mixed types: continuous numerics, categoricals, one binary flag. Cross-sectional, not a time series.

Dataset 2: China Passenger-Vehicle Sales, CPCA (Environment)

Monthly registration records for China's passenger-car market, January 2018–April 2024, from the China Passenger Car Association (CPCA, 2024): 38,806 model-month rows, 10 columns (year-month, make, brand, model, brand country, units sold, EV-versus-gasoline flag, body type, low/high price). Covers 139 brands and 1,345 brand-model combinations (136.9M units); 8,528 EV rows vs 30,278 gasoline, EV unit share 3.3% (2018) → 30.2% (2023). Framed cross-sectionally, as a portrait of the electrification transition.

Dataset 3: Spotify Track Features (Entertainment)

114,000 tracks × 20 columns from the Ultimate Spotify Tracks DB (Hamidani, n.d.): identifiers, a popularity score, and Spotify-computed audio features (danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, tempo, key, mode, time signature, duration, explicit flag) plus a genre label. Artificially balanced at 114 genres × 1,000 tracks.

Dataset 4: TMDB Movie Metadata (Entertainment)

45,466 films × 24 columns from The Movies Dataset (Banik, 2017): budget, revenue, runtime, release date, vote average/count, popularity, language, overview text, tagline, and nested-JSON columns (genres, companies, countries, cast, crew) requiring parsing. The paper's widest mix of data types: numerics, dates, free text, and structured JSON in one table.

Dataset 5: FINRA Equity Short Interest, 30 Sep 2026 (Finance)

The Financial Industry Regulatory Authority's (FINRA, 2026) biweekly equity short-interest file for 30 September 2026 (shrt20260930.csv), used on its own with no price or market-cap data attached: 22,581 stocks × 14 columns. Fields are settlement date, symbol, issuer name, exchange group, market class, current and previous short shares, stock-split flag, average daily volume, days-to-cover, revision flag, percent change, and change in shares. One date, so the table is a cross-section. Short shares range from 1 to 311 million; median days-to-cover 2.4, with FINRA's 999.99 cap on 4,034 names. Days-to-cover is short shares divided by average daily volume. Average daily volume is the only size-like column. The universe is every symbol FINRA reported that day (9,483 OTC, 3,885 Nasdaq, 2,901 NYSE, plus ARCA, BZX, and others), not an S&P 500 slice.

2. Explanation of missing, incorrect, or duplicated data

Dataset 1: SPY Equity Options Chain Snapshot

No missing values. The problem is 625 zero-bid contracts (6.5%), typically deep out-of-the-money long-dated puts quoted 0.00 bid against a wide ask with no volume. These dead quotes cannot be traded; any model trained on them learns from prices that do not exist, so they are discarded outright. A further ~200 ultra-wide-spread contracts (spreads above 50%) are filtered the same way, and implied volatility (range 0.015–4.034, extremes in illiquid names) is winsorized. Volume concentrates in zero-days-to-expiry and weeklies, so analysis is stratified by expiration; mid-prices replace last trades to avoid bid-ask bounce; the Greeks are model-derived features carrying model risk, not ground truth.

Dataset 2: China Passenger-Vehicle Sales, CPCA

Thirty-four exact duplicate (model, year-month) rows are dropped. Body type is the literal string "NA" in 2,366 rows (6.1%), kept as an "Unknown" category since missingness may correlate

with low-volume models. In 6,127 rows (15.8%) the price is 0, dead data that is imputed within brand-model or excluded from price features. The manufacturer field is Chinese while brand is English, so modeling uses brand. February 2020 EV sales collapsed to 8,988 units (COVID lockdown) and the series ends April 2024; structural breaks are handled with regime dummies, never silent deletions.

Dataset 3: Spotify Track Features

450 fully duplicated rows are removed. Subtler: 24,259 track ids repeat because the same song is listed under multiple genres, with identical audio features but different labels, i.e., label noise. Fix: deduplicate to one row per track (most frequent genre) or frame genre as multi-label. The artificial 1,000-tracks-per-genre balance does not reflect listening shares, so prevalence conclusions must be reweighted.

Dataset 4: TMDB Movie Metadata

Seventeen duplicate rows dropped. Many budgets are 0, which is missing data coded as zero; these are recoded as missing and imputed within genre-era cohorts or excluded. Genres, companies, countries, cast, and crew are nested JSON parsed into indicators/counts. Vote averages on few-vote films are shrunk toward the global mean.

Dataset 5: FINRA Equity Short Interest, 30 Sep 2026

No duplicate symbols. One issuer (Nano Labs) has a blank ticker. The split and revision flags are blank unless FINRA applied them (74 splits, 2 revisions). Days-to-cover is floored at 1.00 on 8,295 rows and hits 999.99 on 4,034 thinly traded names. Average daily volume is 0 on 3,143 rows, so days-to-cover is not a real cover time for those names. Percent change versus the prior report is heavy in the tail: 2,546 rows jump over 100% and 843 over 500%; 540 names have prior short interest of 0. These are flagged, not silently deleted.

3. Evaluation and explanation of how and why each dataset is suitable for each of several machine learning methods

Dataset 1: SPY Equity Options Chain Snapshot

Supervised regression: predict implied volatility from strike, moneyness, and time to expiry; each contract carries the quoted IV as target. The smile's nonlinearity suits tree ensembles or a polynomial in moneyness. Classification: liquid vs illiquid from spread, volume, open interest, and distance from spot. Trees fit because the smile is smooth but non-quadratic; the classifier fits because liquidity is a threshold in spread and volume. A deep tree can memorize one day's quotes, so I would hold out expirations (or k-fold by tenor) and cap depth so the smile generalizes to unseen strikes.

Dataset 2: China Passenger-Vehicle Sales, CPCA

Classification: predict the EV-versus-gasoline flag from brand, brand country, body type, and price band. Tree ensembles suit the mixed categorical features without a one-hot explosion. Regression: predict monthly EV units from lagged sales and segment mix. Expanding-window splits (no leakage) and a seasonal baseline are the overfitting guards: a booster that only looks good inside 2021-2022 has not generalized.

Dataset 3: Spotify Track Features

Classification: predict popularity tier (or genre, post-dedupe) from audio features, to see which sonic attributes the market rewards. Clustering: group tracks into data-derived archetypes (high-energy/low-acousticness) for playlist design; the features are dense and continuous, which suits clustering. Split by track id (grouped k-fold), not by row, so the same song cannot sit in train and test under two genre labels.

Dataset 4: TMDB Movie Metadata

Regression: predict log box-office revenue from budget, genre, cast/crew, timing, and pre-release popularity, the studio greenlight problem. Classification: hit vs flop (revenue > 2x budget) from pre-release features only, evaluated with precision-recall for the asymmetric cost of a false greenlight. A year-based hold-out plus a regularized log-linear model, or a depth-capped tree, keeps a 1995 film from training the 2016 greenlight.

Dataset 5: FINRA Equity Short Interest, 30 Sep 2026

Regression: predict log short shares from log average daily volume and market class, dropping the 3,143 zero-volume names (R-squared 0.42; log correlation 0.65). The residual versus that volume prediction is the anomaly score. Classification: crowded = days-to-cover above 5 (37% of rows), predicted from volume quartile and market class (48% in the lowest-volume quartile, 25% in the highest); short shares and days-to-cover stay out of the features. Clustering on log volume, log short shares, and days-to-cover can separate crowded OTC names from liquid large caps. One row per symbol, so a random split is valid; I would stratify by market class, keep the tree shallow, and treat a collapse to the majority class on the hold-out as overfitting to the OTC days-to-cover cap.

4. Industry and problem: explanation of the kinds of questions that could be answered by each dataset

Dataset 1: SPY Equity Options Chain Snapshot (Finance: derivatives)

Can ML recover the volatility smile by predicting IV from moneyness and expiry? Matters to volatility trading and market-making; personally, as a valuation specialist at a Hong Kong hedge fund, I want to prove out the volatility smile with data science methods rather than take it as a textbook given: an empirical test of the constant-volatility assumption behind Black-Scholes (Black & Scholes, 1973). Do market Greeks behave as textbooks say across moneyness and

tenor? I verified each behavior empirically here, useful for valuation and risk management and as a check on the data's consistency. What drives IV most: moneyness, expiry, or liquidity? Tells pricing/risk functions where modeling effort belongs.

Dataset 2: China Passenger-Vehicle Sales, CPCA (Environment: automotive)

Which attributes best predicted electrification: price segment, body type, or brand origin? Matters to auto strategy; personally, the demand side of emissions. Did the January 2023 subsidy end change EV price elasticity? A natural policy experiment in official-registration data. Can monthly EV sales be forecast from lags and mix, and where does a seasonal baseline break in 2021–2022? Matters to inventory and battery-supply planning.

Dataset 3: Spotify Track Features (Entertainment: music)

Which audio features predict popularity: energy, danceability, valence? Matters to streaming economics and A&R; personally, a listener's curiosity. Do features cluster into stable archetypes for playlist design? Matters to recommendation, where the task is matching songs to moods.

Dataset 4: TMDB Movie Metadata (Entertainment: film)

What drives revenue: budget, genre, cast, timing, pre-release popularity? Matters to greenlighting; personally, what the market pays for as a film consumer. Can a pre-release hit/flop classifier flag failures before money is spent? Matters to risk management where one flop erases a slate.

Dataset 5: FINRA Equity Short Interest, 30 Sep 2026 (Finance: equities)

Which names are heavily shorted relative to how much they trade, and are they the lower-volume names? Matters to risk managers, prime brokers, and securities lenders: a large short position in a thinly traded stock is harder to cover and carries more squeeze risk, and a lender sees more borrow demand. Personally relevant as a hedge-fund professional. Does counting shares or

measuring days-to-cover change who looks crowded? Only 1 of the 50 largest short positions is also among the 50 with the highest days-to-cover. The largest share counts are Nvidia, Transocean, Plug Power, Grab, and Opendoor; the highest days-to-cover are illiquid OTC names at FINRA's 999.99 cap. Are large percent changes real shorting or corporate actions such as splits? Matters to anyone building a short-interest monitor without false alarms.

5. Summary of your experiences with the project to this point

The project started broad and narrowed to five datasets I could defend in finance, environment, and entertainment. Week 1 put the business problem before the dataset; Week 2's CRISP-DM framing made each candidate earn a prediction task. The SPY snapshot was the turning point: IV against strike reproduced the skew. Weeks 3-4 treated the chain as supervised segmentation. Overfitting is when a model learns the noise in its training data instead of the real pattern, so it scores great on the data it trained on but fails on new data. Memorization versus understanding. Fitting a volatility smile so tightly to one day's quotes that it nails every print but blows up the next day is what that looks like in my work. That is why the course pairs it with train/test splits and cross-validation. The cure is always testing on data the model has not seen.

Zero-bid contracts, duplicate rows, zero-coded budgets, and corporate-action jumps looked like footnotes until I asked what each one does to the target. That is Week 5 in practice: a model is only as honest as the labels it learns from. This paper treats each file as its own table. In my opinion, joining datasets is an important next step. It enriches the data enough to address real-world issues a single table cannot: short interest without prices, a smile without realized volatility, sales without a policy calendar.

Appendix A: Citations

Banik, R. (2017). The movies dataset [Data set]. Kaggle. Metadata compiled from The Movie Database (TMDB) API and GroupLens MovieLens. https://www.kaggle.com/datasets/rounakbanik/the-movies-dataset

Black, F., & Scholes, M. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy, 81*(3), 637–654. https://doi.org/10.1086/260062

China Passenger Car Association. (2024). *Monthly passenger-vehicle sales records, January 2018–April 2024* [Data set]. Kaggle. Compiled by Felix Zhao. https://www.kaggle.com/datasets/felixzhao/china-automobile-monthly-sales-data-2018-2024-4

Dubach, P. (2025). *Historic options dataset: SPY, IWM, and QQQ options 2008–2025* [Data set]. GitHub. SPY options chain, December 31, 2024. https://github.com/philippdubach/historic-options-dataset

Financial Industry Regulatory Authority. (2026). *Equity short interest* [Data set]. Biweekly file shrt20260930.csv, settlement date 30 September 2026, all reported symbols. https://cdn.finra.org/equity/otcmarket/biweekly/shrt20260930.csv

Hamidani, Z. (n.d.). *Ultimate Spotify tracks DB* [Data set]. Kaggle. Audio features computed by the Spotify Web API. https://www.kaggle.com/datasets/zaheenhamidani/ultimate-spotify-tracks-db

Appendix B: AI Appendix

I used AI to brainstorm topic areas and to identify datasets that met my requirements (at least 10,000 rows and a sufficient mix of column types). I also used AI to fit the document to the

required formatting: 12-point Times New Roman, double spacing, one-inch margins, and the rubric’s section labels.

Appendix C: Dataset Examples (first three rows)

Each table shows the first three rows of the dataset file, restricted to key columns for page width, except Dataset 1: its first rows are deep in-the-money contracts expiring December 31, 2024, so three consecutive near-the-money contracts from the file are shown instead. Full column lists appear in Section 1.

Dataset 1: SPY options chain snapshot, near-the-money contracts (6 of 23 columns)

contract_id	strike	type	last	volume	implied_volatility
SPY250207C00594000	594.0	call	7.8	275	0.13195
SPY250207P00594000	594.0	put	12.94	12	0.1417
SPY250207C00595000	595.0	call	7.39	134	0.13195

Dataset 2: China passenger-vehicle sales (5 of 10 columns)

year_month	brand	model	units_sold	is_ev
2022-11-01	Tesla	Model Y	69098	EV
2022-11-01	Wuling Motors	宏光 MINIEV	68567	EV
2022-11-01	BYD	宋 PLUS 新能源	64145	EV

Dataset 3: Spotify track features (5 of 20 columns)

artists	track_name	popularity	danceability	track_genre
Gen Hoshino	Comedy	73	0.676	acoustic
Ben Woodward	Ghost - Acoustic	55	0.42	acoustic
Ingrid Michaelson;ZAYN	To Begin Again	57	0.438	acoustic

Dataset 4: TMDB movie metadata (6 of 24 columns)

budget	release_date	revenue	title	vote_average	vote_count
30000000	1995-10-30	373554033	Toy Story	7.7	5415
65000000	1995-12-15	262797249	Jumanji	6.9	2413
0	1995-12-22	0	Grumpier Old Men	6.5	92

Dataset 5: FINRA equity short interest, 30 Sep 2026 (6 of 14 columns)

settlementDate	symbolCode	currentShortPositionQuantity	previousShortPositionQuantity	changePercent	daysToCoverQuantity
2026-09-30	A	5,240,309	5,078,629	3.18	1.78
2026-09-30	AA	11,725,447	11,984,650	-2.16	2.85
2026-09-30	AAA	7,957	1,429	456.82	1.00